

DP Computer Science: A 1.2.1–1.2.2 Representing Data (Combined Overview)

1. Overview

This section explores how computers represent data. It covers:

- **A1.2.1:** Number systems (binary, hexadecimal, decimal) and conversion methods
- **A1.2.2:** How computers use binary to store various data types—integers, text, images, audio, and video

2. A1.2.1 — Number Systems

Learning Intentions (A1.2.1)

By the end of this section, students will be able to:

1. Explain the concept of different number systems (binary, hexadecimal, decimal) and their significance in computing
2. Identify the base of each number system and the range of digits used
3. Explain the relationship between place values and powers of the base in each number system
4. Represent positive integers in binary and hexadecimal notation
5. Convert binary integers to decimal and vice versa
6. Convert hexadecimal integers to decimal and vice versa
7. Convert directly between binary and hexadecimal using four-bit grouping

Key Vocabulary (A1.2.1)

Term	Definition
Bit	A single binary digit, either 0 or 1
Nibble	A group of four bits, which is exactly one hexadecimal digit
Byte	A group of eight bits
Base (radix)	The number of digit symbols a system uses: 2 for binary, 16 for hexadecimal, 10 for decimal
Place value	The value of a position, each one a fixed multiple of the position to its right
Hexadecimal	A base-16 system using 0–9 then A–F, where one digit stands for four bits

Number Systems Summary

Feature	Binary	Decimal	Hexadecimal
Base	2	10	16
Digits	0, 1	0–9	0–9, A–F
Place Values	Powers of 2	Powers of 10	Powers of 16

Feature	Binary	Decimal	Hexadecimal
Example	1010 ₂	10 ₁₀	A ₁₆
Use in Computing	Machine-level data	Human-readable	Compact binary representation

Place-Value Grids

Binary Place Values:

128	64	32	16	8	4	2	1
2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

Hexadecimal Place Values:

256s	16s	1s
16 ²	16 ¹	16 ⁰

Conversion Methods

Binary ↔ Decimal

Binary → Decimal (Place-Value Method):

Binary	1	0	1	1	0	1
Place Value	32	16	8	4	2	1
Calculation	32	0	8	4	0	1
Sum						45

Decimal → Binary (Repeated Division):

Step	Division	Quotient	Remainder
1	45 ÷ 2	22	1
2	22 ÷ 2	11	0
3	11 ÷ 2	5	1
4	5 ÷ 2	2	1
5	2 ÷ 2	1	0
6	1 ÷ 2	0	1

Read remainders from bottom to top: 101101₂

Hexadecimal ↔ Decimal

Hexadecimal → Decimal:

Hex	2	F
Place Value	16	1
Calculation	$2 \times 16 = 32$	$15 \times 1 = 15$
Sum	47 ₁₀	

Decimal → Hexadecimal:

Step	Division	Quotient	Remainder
1	$206 \div 16$	12	14 (E)
2	$12 \div 16$	0	12 (C)

Read remainders from bottom to top: CE₁₆

Binary → Hexadecimal (Four-Bit Grouping)

Binary → Hexadecimal:

Step	Example
1. Group bits from the right	11011110 ₂ → 1101 1110
2. Convert each nibble	1101 ₂ = D, 1110 ₂ = E
3. Result	DE ₁₆

Hexadecimal → Binary:

Step	Example
1. Convert each hex digit to 4 bits	3A ₁₆ → 0011 1010
2. Remove leading zeros if desired	111010 ₂

Hexadecimal to Binary Reference Table

Hex	Binary	Hex	Binary
0	0000	8	1000
1	0001	9	1001
2	0010	A	1010
3	0011	B	1011
4	0100	C	1100
5	0101	D	1101
6	0110	E	1110
7	0111	F	1111

3. A1.2.2 — Binary Data Storage

Learning Intentions (A1.2.2)

By the end of this section, students will be able to:

1. Explain the concept of **binary encoding** and how it uses only two states (0 and 1) to represent all data
 2. Describe the relationship between binary code and the **on/off states** of electrical signals in a computer
 3. Explain the impact of binary encoding on **data storage, retrieval, and processing**
 4. Describe how various **data types** are stored in binary form:
 - **Integers** — positive integers represented directly using binary code
 - **Text/Characters** — encoded using ASCII and Unicode
 - **Images** — grid of pixels, each pixel's colour encoded as binary value(s)
 - **Audio and Video** — digitised by sampling analogue signals
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Key Vocabulary (A1.2.2)

Term	Definition
Binary encoding	Representing all data using only two states (0 and 1)
ASCII	American Standard Code for Information Interchange; 7-bit character encoding
Unicode	Universal character encoding standard; supports characters from all writing systems
Pixel	The smallest unit of a digital image; each pixel stores colour as binary values
Sampling	Converting continuous analogue signals into discrete digital values
Digitisation	The process of converting analogue information into digital (binary) form

Data Type Storage Mechanisms

1. Integers

Aspect	Description
Method	Positive integers represented directly using binary (unsigned)
Example	$45_{10} = 00101101_2$ (8-bit representation)
Limitation	Unsigned binary cannot represent negative numbers; signed representations (two's complement) are used for negative values

2. Characters and Text

Standard	Description
ASCII	7-bit encoding; 128 characters (English letters, digits, punctuation)
Unicode	16-bit+ encoding; supports 143,000+ characters across all writing systems

ASCII Example:

Character	ASCII Code (Decimal)	Binary
A	65	01000001
B	66	01000010
a	97	01100001
0	48	00110000

Why Unicode Replaced ASCII:

- ASCII insufficient for non-English languages
- Unicode supports **all** characters from **all** writing systems
- UTF-8/UTF-16 provide efficient encoding for web and software

3. Images

Aspect	Description
Structure	Grid of pixels (picture elements)
Colour Encoding	Each pixel's colour encoded as binary value(s)
Grayscale	One value per pixel (e.g., 8-bit = 256 shades of grey)
RGB Colour	Three values per pixel: Red, Green, Blue (e.g., 24-bit = 16.7 million colours)

Example — 3×3 Grayscale Image:

255	0	128
64	192	32
0	255	200

Each value is stored as an 8-bit binary number.

4. Audio

Aspect	Description
Process	Analogue sound waves are sampled at regular intervals
Sample	Each sample measures amplitude at a point in time
Quantisation	Amplitude values are rounded to discrete binary numbers

Aspect	Description
File Formats	WAV, MP3, AAC

Example — Digitisation:

text

Analogue Signal: [continuous wave]

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Sampling: [discrete points at intervals]

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Quantisation: [amplitude values rounded]

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Binary: [10110011 01100110 ...]

5. Video

Aspect	Description
Structure	Sequence of frames (individual images)
Framerate	Frames per second (e.g., 24, 30, 60 fps)
Encoding	Each frame stored as image; compression reduces file size
File Formats	MP4, AVI, MOV

Binary Encoding Summary

Data Type	Encoding Method	Key Feature
Integers	Binary (place value)	Direct binary representation
Text	ASCII / Unicode	Each character assigned a unique code
Images	Pixel grid	Each pixel colour = binary value(s)
Audio	Sampling + quantisation	Amplitude values as binary
Video	Sequence of frames	Frames stored as compressed images

4. Example Hinge Questions

Question	Purpose
"Why is it necessary to convert information into discrete units for computers to process it?"	Checks understanding of digital data
"Can you give an example of how binary code might be used in a real-world application, besides storing numbers?"	Encourages thinking beyond simple number representation

Question	Purpose
"What are the key differences between decimal, binary, and hexadecimal? Why do we need all three?"	Checks comprehension of purpose and characteristics
"What happens if we try to represent a negative number using only unsigned binary?"	Prompts thinking about limitations and signed representations
"Why is hexadecimal useful when working with binary data?"	Assesses understanding of hexadecimal's role
"Why was ASCII insufficient for representing text in modern computing?"	Checks understanding of ASCII limitations and Unicode advantages
"How does the binary representation of an image differ from that of a piece of text?"	Encourages comparison of different data types

5. Differentiation

Level	Strategy
Support	Provide pre-drawn place-value grids; start with values under 16; offer step lists for repeated-division method
Challenge	Two-byte conversions; spot-the-error tasks; research different encoding schemes (ASCII, Unicode, image formats, audio formats)

6. Resources

- [Presentation \(Slide Deck\)](#)
- [InThinking eBook \(A1.2.1–A1.2.2 with quiz\)](#)
- [Worksheet: Number system conversions + data types](#)
- [Worksheet Answers](#)

7. ATL Skills Summary

Skill	Description
Research	Investigate different data encoding schemes and their applications
Thinking	Apply critical thinking to analyse how different data types are represented and stored
Communication	Clearly explain the process of binary encoding and its impact on data storage and retrieval
Self-Management	Monitor own conversion accuracy; choose reliable method

8. Key Takeaway

All data stored in a computer is represented using binary (0s and 1s). Different data types—numbers, text, images, audio, video—use different encoding schemes to map information to binary values, but the underlying storage is always binary.